



Adversary Emulation Practice

1. Introduction

Adversary emulation is a proactive cybersecurity practice that simulates real-world attacker behaviors to evaluate an organization's detection and response capabilities. Unlike vulnerability scanning, adversary emulation focuses on replicating attacker tactics, techniques, and procedures (TTPs) to validate SOC readiness.

This exercise used **MITRE Corporation Caldera** to simulate adversary behavior and **Wazuh Inc. Wazuh** to validate detection capabilities.

The simulated attack mapped to **MITRE ATT&CK Technique T1566 – Phishing**.

2. Objective

The objectives of this task were to:

- Simulate a spearphishing attack.
 - Monitor detection capabilities in Wazuh.
 - Validate alert generation accuracy.
 - Identify detection gaps.
 - Improve defensive configurations.
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3. Theoretical

3.1 What is Adversary Emulation?

Adversary emulation replicates known attacker behaviors based on threat intelligence. It differs from penetration testing because it focuses on realistic attack chains aligned with MITRE ATT&CK techniques.

Benefits include:

- Testing detection rules.
 - Evaluating SOC response.
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- Identifying monitoring blind spots.
- Improving alert tuning.

3.2 MITRE ATT&CK Mapping

This exercise focused on: **T1566 – Phishing**

Description: Adversaries send fraudulent emails to trick users into revealing credentials or executing malicious payloads.

4. Practical Implementation

4.1 Emulation Environment

- Windows Virtual Machine (Target)
- MITRE Caldera (Attack Simulation)
- Wazuh Agent installed on endpoint
- SIEM dashboard configured

4.2 Emulation Execution

Using Caldera:

1. Selected phishing ability from ATT&CK profile.
2. Simulated email-based payload execution.
3. Triggered PowerShell-based malicious command.

The payload execution generated endpoint artifacts.

5. Detection Validation in Wazuh

Wazuh successfully generated alerts related to:

- Suspicious PowerShell execution
 - Unusual script execution behavior
 - Potential phishing activity indicators
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Detection Log Summary

Timestamp	TTP	Detection Status	Notes
2025-08-18 17:00:00	T1566	Detected	Suspicious email activity
2025-08-18 17:02:00	T1059	Detected	PowerShell execution

T1059 (Command and Scripting Interpreter) was triggered due to PowerShell usage during simulation.

6. Detection Analysis

The SOC detection system demonstrated:

- Successful identification of suspicious script execution.
- Alert generation within minutes.
- Accurate mapping to known attack techniques.
- Proper log correlation between endpoint and SIEM.

No critical detection failure was observed.

7. Detection Gaps Identified

Although detection was successful, the following improvements were noted:

- Email metadata logging was limited.
- No automatic attachment sandboxing.
- No user behavioral anomaly detection.
- No automated endpoint isolation triggered.

These represent areas for SOC maturity enhancement.

8. Emulation Report Summary

An adversary emulation exercise was conducted using MITRE Caldera to simulate a phishing-based attack mapped to MITRE ATT&CK technique T1566. The simulated payload execution triggered suspicious PowerShell activity on the target system. Wazuh successfully detected abnormal script execution and generated alerts within minutes. The SOC monitoring



system demonstrated effective detection capability for scripting-based threats. However, improvements are recommended in email metadata monitoring and automated containment processes. This exercise validated detection controls while highlighting opportunities to enhance proactive monitoring and automation within the SOC environment.

9. Recommendations

- Enable advanced email security logging.
 - Deploy sandbox analysis for attachments.
 - Implement automated endpoint isolation for high-risk alerts.
 - Expand detection rules for credential harvesting behavior.
 - Conduct regular adversary emulation exercises.
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10. Key Learnings

- Adversary emulation validates real detection capabilities.
 - Mapping alerts to MITRE ATT&CK improves clarity.
 - Continuous testing strengthens SOC resilience.
 - Detection tuning is an ongoing process.
 - Proactive validation reduces real-world attack impact.
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11. Conclusion

This adversary emulation exercise successfully demonstrated the effectiveness of SOC detection capabilities against phishing-based attack simulations. By leveraging MITRE Caldera for emulation and Wazuh for monitoring, the organization validated its ability to detect suspicious scripting activity. Continuous adversary simulation enhances security posture and ensures that defensive mechanisms remain effective against evolving threats.